Properties of Simple Molecular Compounds

Exam code: 0620 & 0971

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Properties of simple molecular compounds

- Small molecules are compounds made up of molecules that contain just a few atoms **covalently bonded** together
- They have **low** melting and boiling points so covalent compounds are usually **liquids** or **gases** at room temperature
- As the molecules increase in **size**, the melting and boiling points generally increase
- Small molecules have poor electrical conductivity

Explaining the properties of simple molecular compounds

Extended tier only

Linking bonding and properties

- Small molecules have covalent bonds joining the atoms together, but weak intermolecular forces that act

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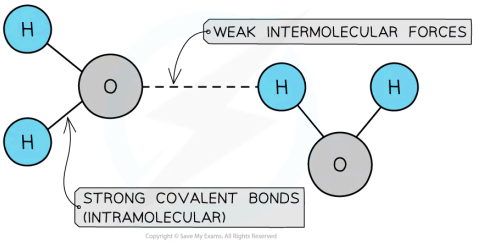
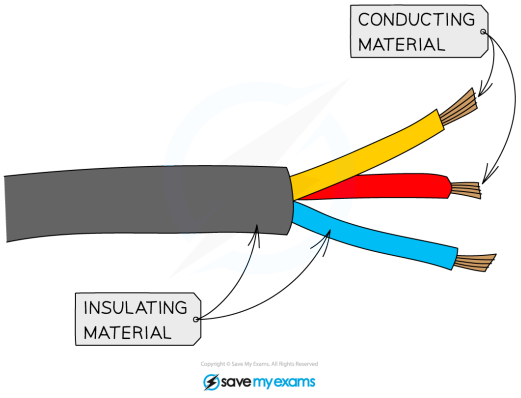
molecules
- These forces are **very weak** when compared to the covalent bonds and so most small molecules are either gases or liquids at room temperature
- As the molecules increase in **size** the intermolecular forces also increase as there are more electrons available
- This causes the melting and boiling points to **increase**



The bonds between hydrogen and oxygen in water are COVALENT, and the attractions between the molecules are INTERMOLECULAR FORCES which are about one tenth as strong as covalent bonds

Electrical Conductivity

- Molecular compounds are **poor conductors** of electricity as there are no free ions or electrons to carry the charge.
- Most covalent compounds do not conduct at all in the solid state and are thus **insulators**
- Common insulators include the plastic coating around household electrical wiring, rubber and wood



The plastic coating around electrical wires is made from covalent molecules that do not allow a flow of charge

Examiner Tips and Tricks

The atoms within covalent molecules are held together by covalent bonds while the molecules in a covalent substance are attracted to each other by intermolecular forces.

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